

PROJECT REPORT

Project Title: Predictive Modeling for Course Demand and Revenue Forecasting

Domain: Data Analytics & Machine Learning

Platform: EduPro Online Platform

1. Executive Summary

This project builds an end-to-end Machine Learning solution to forecast course enrollments and revenue for the EduPro platform. By integrating raw relational data across four distinct sheets (Users, Teachers, Courses, and Transactions), the pipeline cleanses, transforms, and engineers features to train a Random Forest Regressor model, deployed via an interactive Streamlit web dashboard.

2. Problem Statement

Online education platforms struggle to optimize pricing strategies and instructor allocations due to unpredictable course demand. The objective is to accurately forecast total student enrollments and expected financial revenue for new or existing courses using historic platform metrics.

3. Dataset Architecture & Preprocessing

Data Sources: Merged 4 relational datasets (Users, Teachers, Courses, Transactions).

Feature Engineering: Calculated total transactions and revenue per course via aggregate grouping.

Encoding: Applied One-Hot Encoding on categorical attributes (CourseCategory, CourseType, CourseLevel).

4. Machine Learning Implementation

Algorithm: Random Forest Regressor

Target Variables: Total_Enrollments and Total_Revenue

Performance Metrics: Evaluated model predictions using R^2 Score ($R^2 = 0.96$ for Revenue Forecasting).

5. Interactive Deployment

Built a web dashboard using Streamlit allowing business stakeholders to adjust course parameters (Price, Duration, Rating, Teacher Experience) in real-time and observe predicted user demand and revenue outputs dynamically.



